RAIKES SCHOOL

University of Nebraska Lincoln

Decision-Making and Reinforcement Learning

Management Science Principles

Motivation

- Given constraints or a fixed system, we can use optimization to make a decision once.
  - Everything from setting model parameters to supply chain logistics

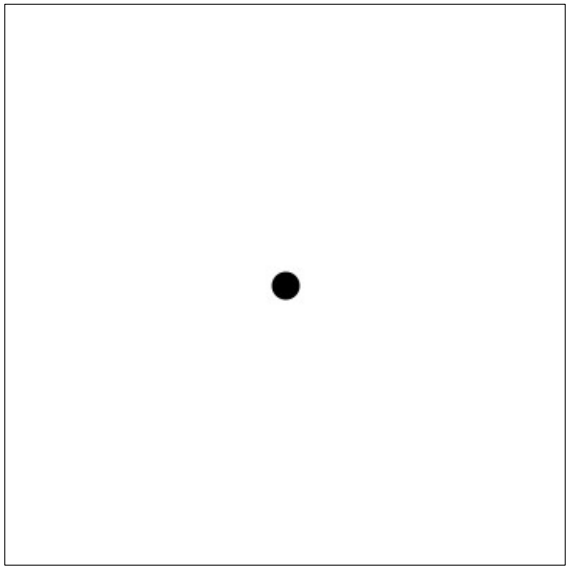
- Evolutionary algorithms included an element of simulation (through generations) and rules (fitness functions) to search an unpredictable space for optimal values.

- Let’s consider simulations as they pertain to making decisions.

Random Walk

Random Walk

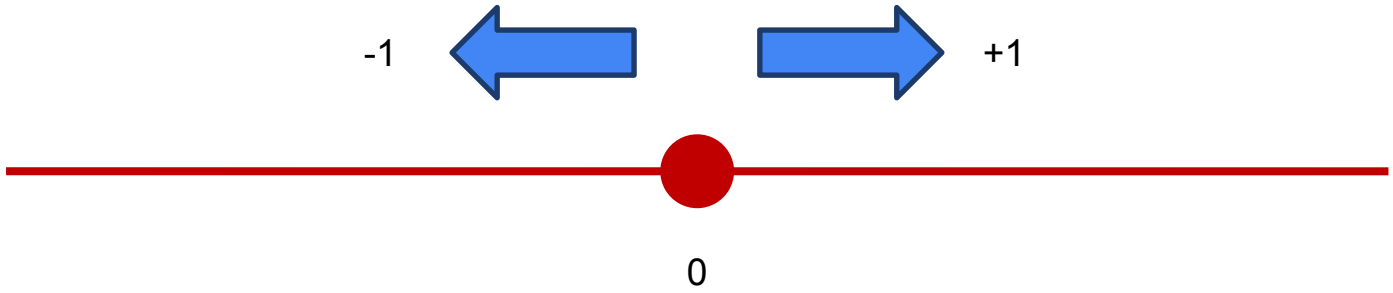
- We have already seen Monte Carlo methods broadly as simulated runs of an experiment to determine outcomes.
- Random walk is one such algorithm.



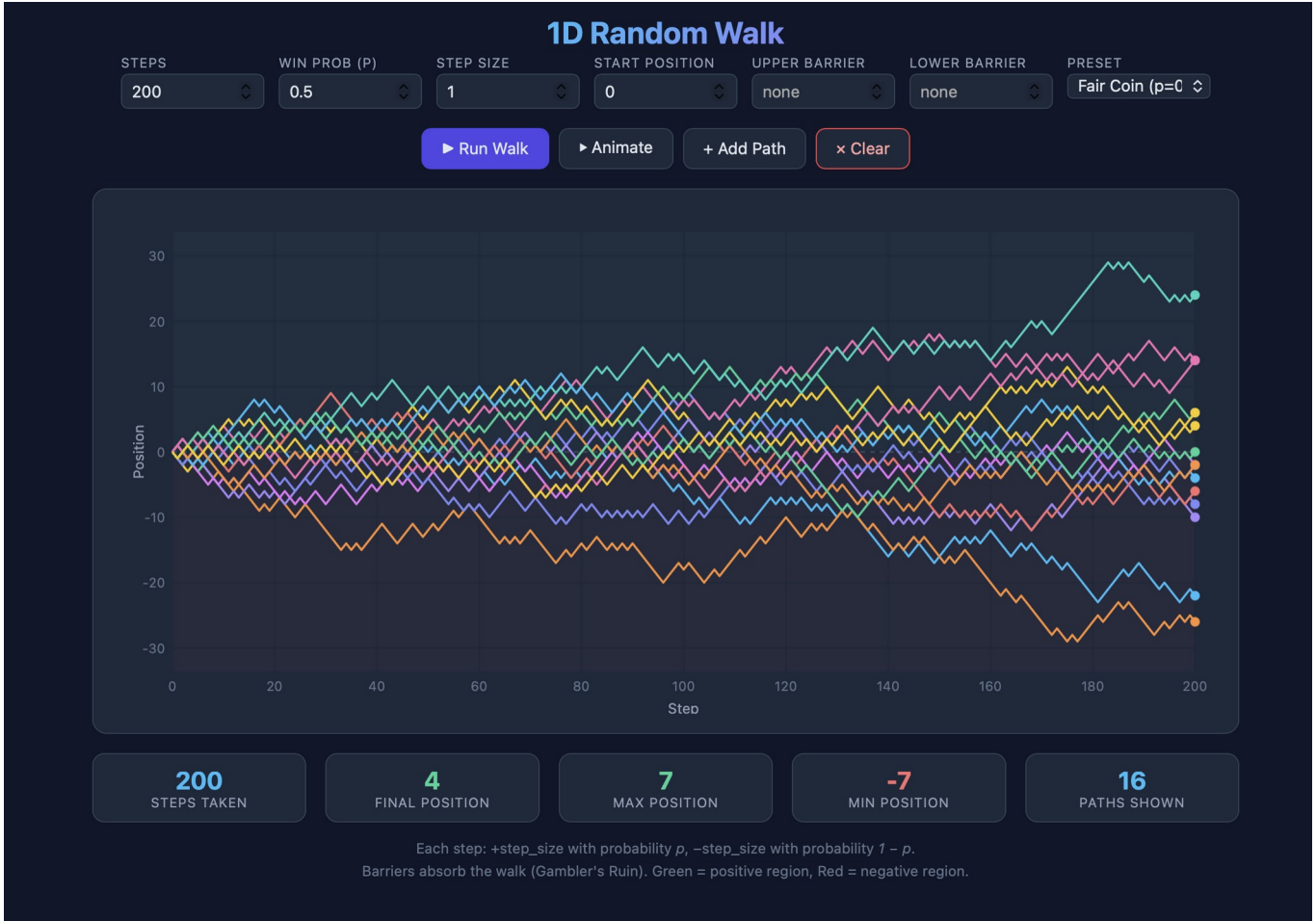
Credit: https://rdlyons.pages.iu.edu/rw/rw.html

1D Random Walk

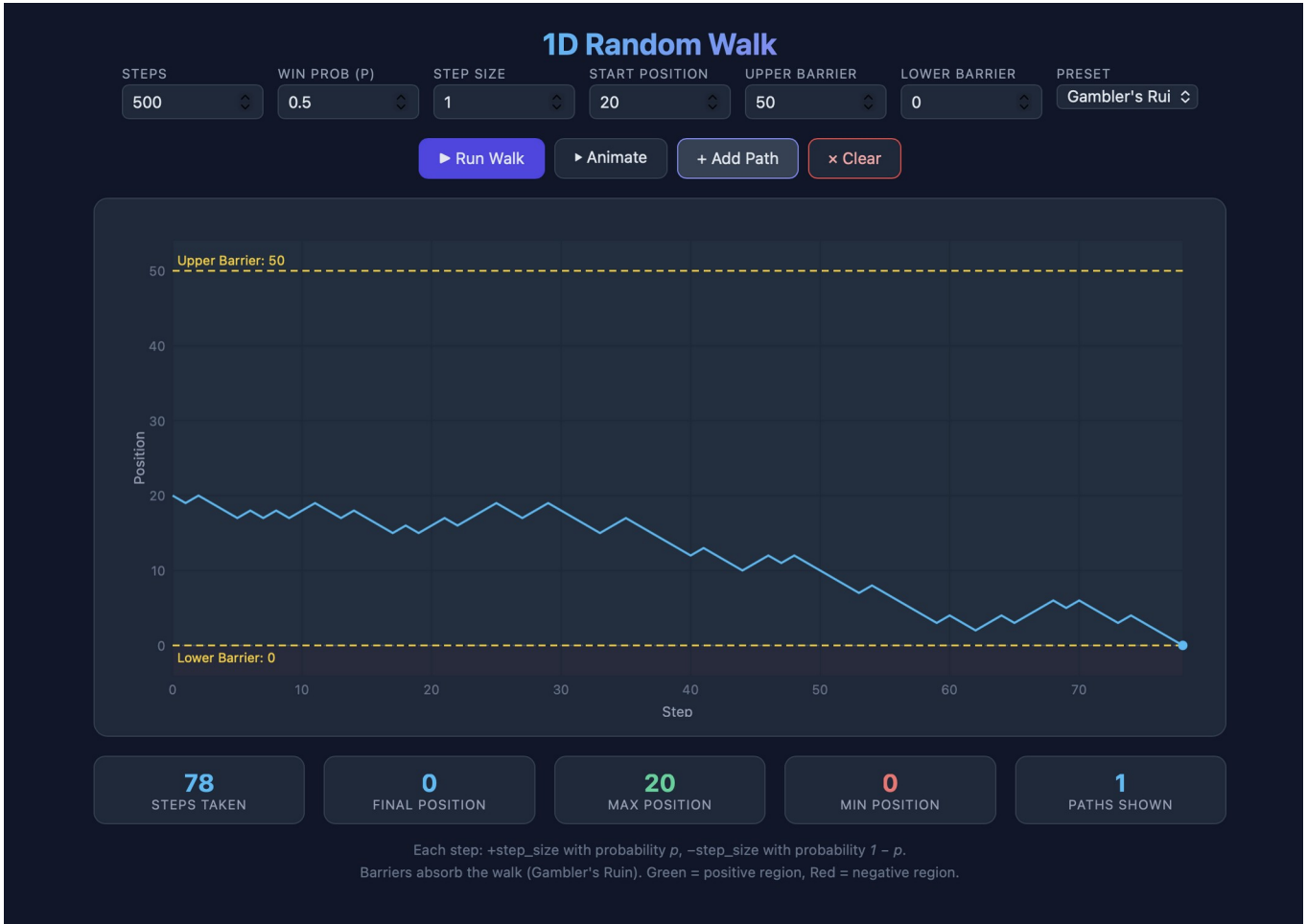
- Consider a scenario where the event can either go left or right: -1 or +1
- We can model this as a 1-dimensional random walk.
- The randomness of the walk direction can be controlled as part of the simulation.



The benefit of simulation is visualizing many sequences or paths



For instance, visualizing minimum thresholds…



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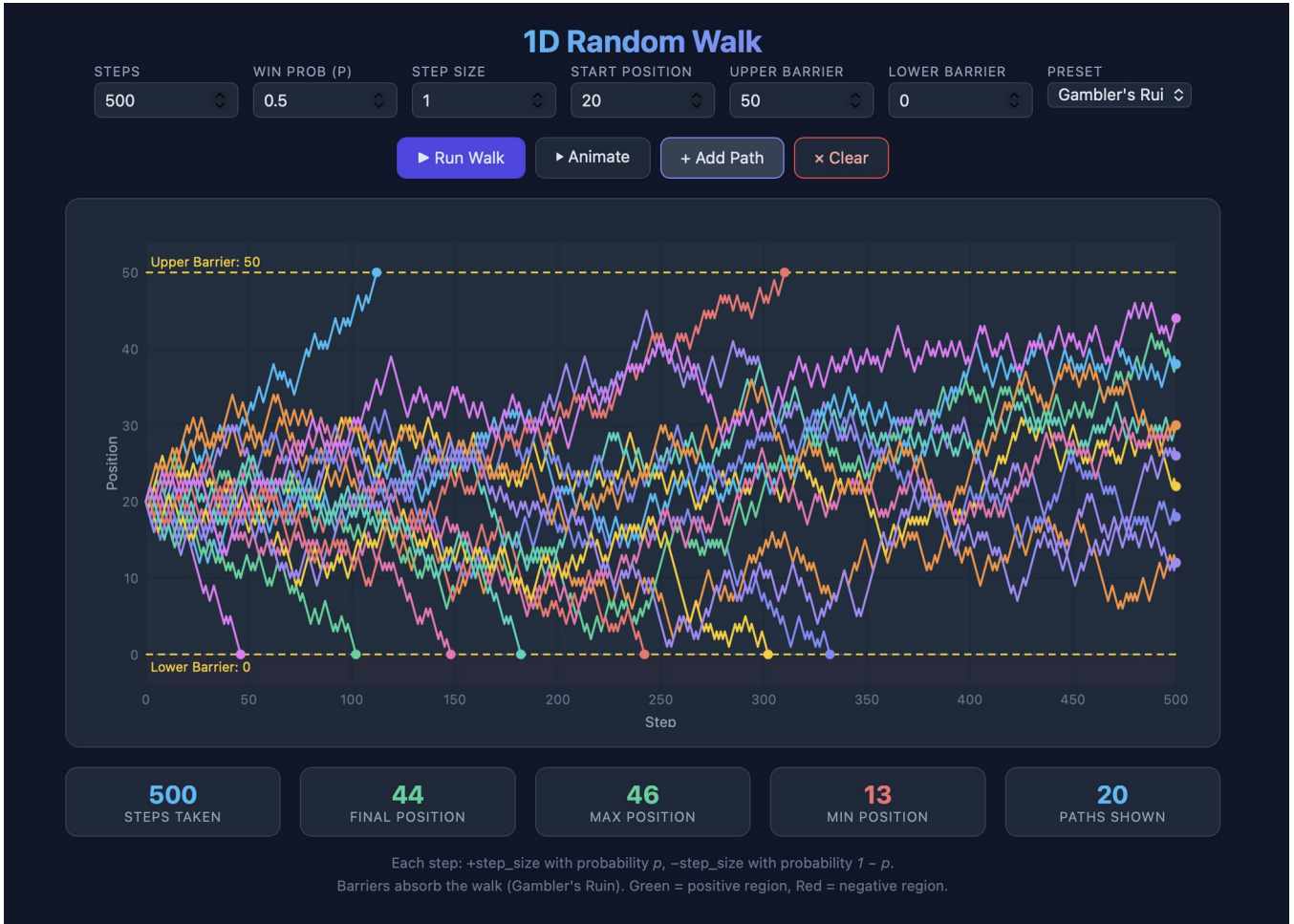
… over many iterations



And seeing how even a small change in probability can have major impacts

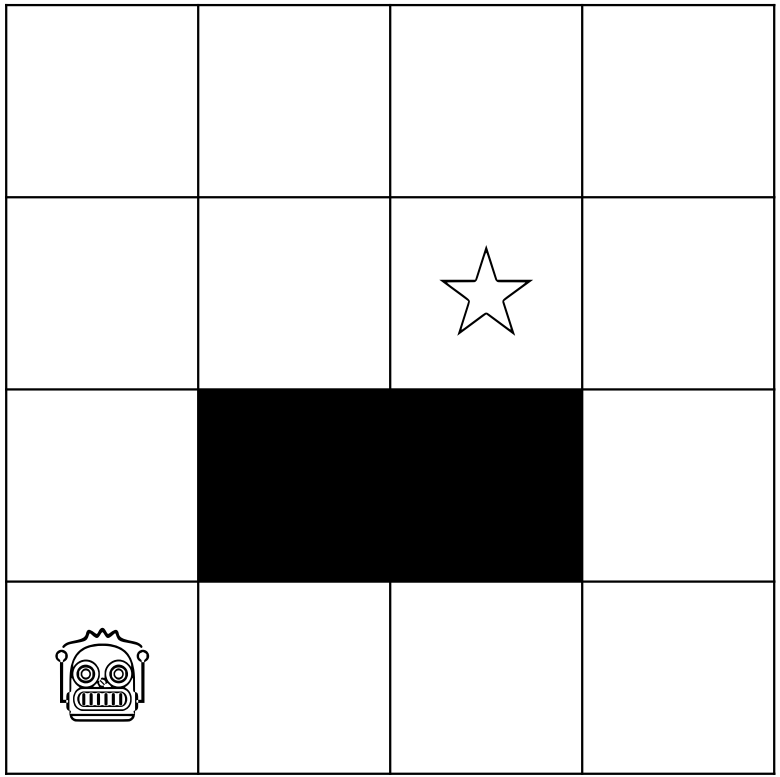
Supporting Dynamicism

- Even simple stochastic systems can produce surprising behavior, and simulations help us see it.
- But… these are still modeling static systems, using a fixed probability across all trials.
  - The only sequential aspect is starting from the point of the last iterations, like a walk where each step is an iteration.
- Consider the gambling scenario: a person may adopt a series of heuristics:
  - If bankroll < $10, bet smaller
  - If on a winning streak, increase bets
  - Quit after 100 rounds
- Such heuristics are common in human decision making with many business parallels:
  - Thresholds at which to reorder inventory
  - Cut ad spend if ROI drops below target
  - Raise prices during peak demand

Let’s try a machine learning approach to selecting rules!

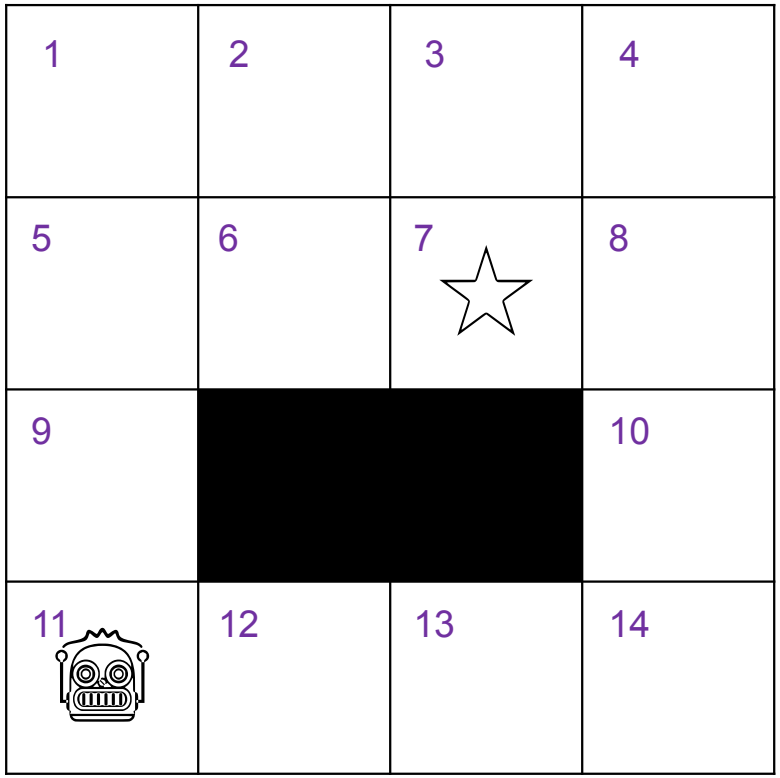
Reinforcement Learning

Reinforcement Learning

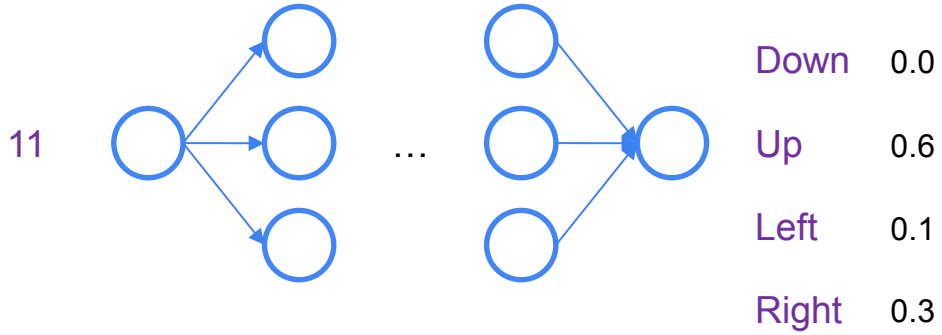


- Broadly, RL models the algorithm as an agent inside the environment.
- The agent should take **actions** that give it a **reward** or move it closer to the goal.
- RL includes **state** as a way to measure action efficacy and seek rewards.

The Policy Network



- Let's construct a simple model to control the actor, call this the Policy or Actor network.
- This network should take as input the current **State** and output an **Action**.
- Here, we could simply encode state as location, but you can imagine more sophisticated state encodings.



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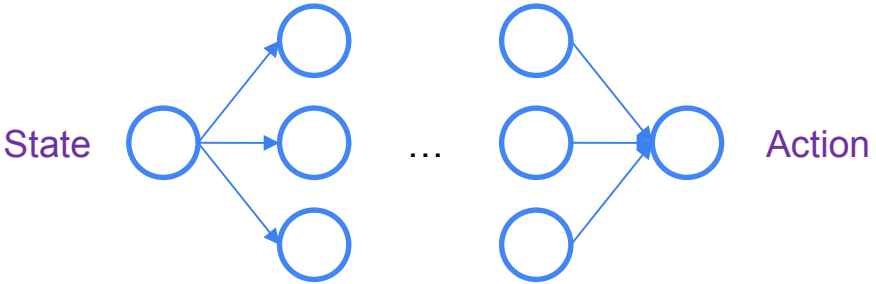
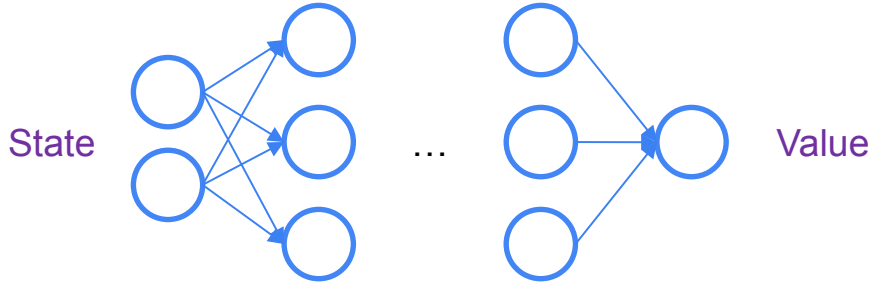
The Value Network

- How can we train the weights for this policy network?
- Modern RL usually creates an adversarial architecture:
  - A Critic or Value network will take an action and state and try to predict how *good* it is.
    - Depending on the technique being used, you may hear this called the value, the fit, the reward, or the “quality” of an action.
      - For instance, Deep Q-Learning (DQL) is concerned with building deep networks that can predict future reward in high-dimensional spaces like entire video game worlds by looking at the pixel values!

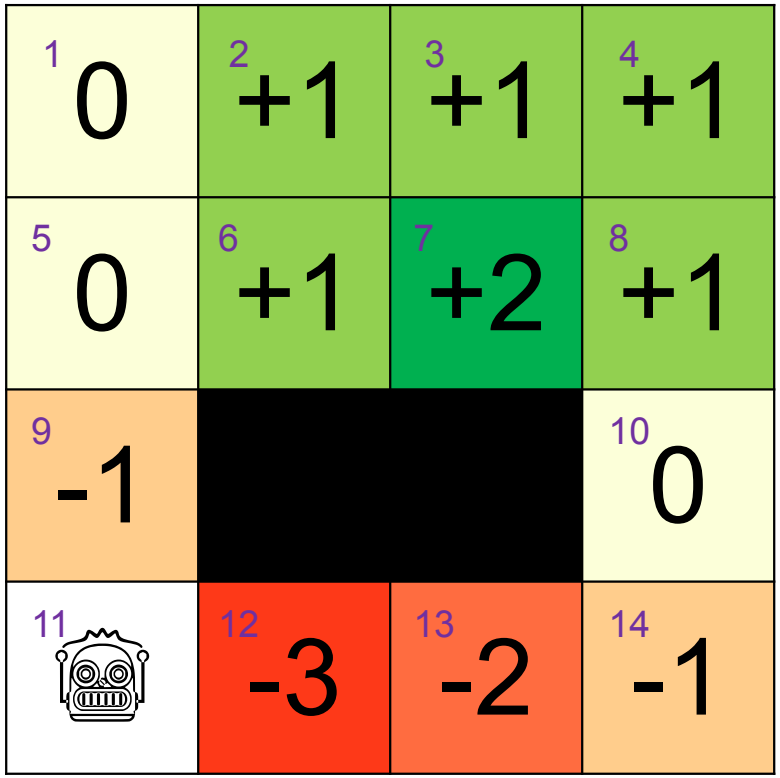
Policy network



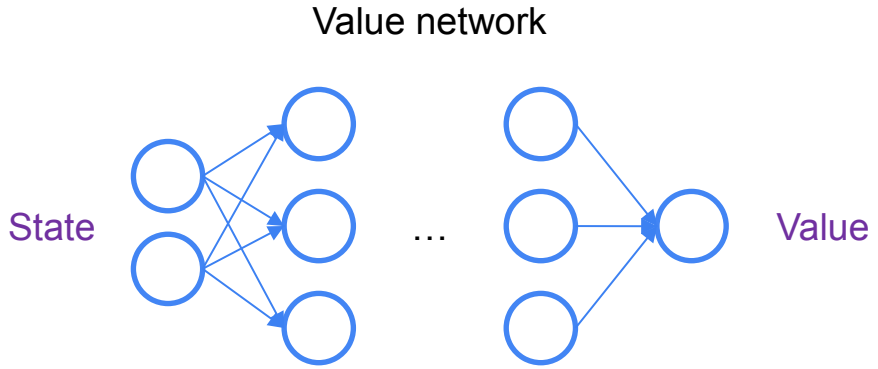
Value network



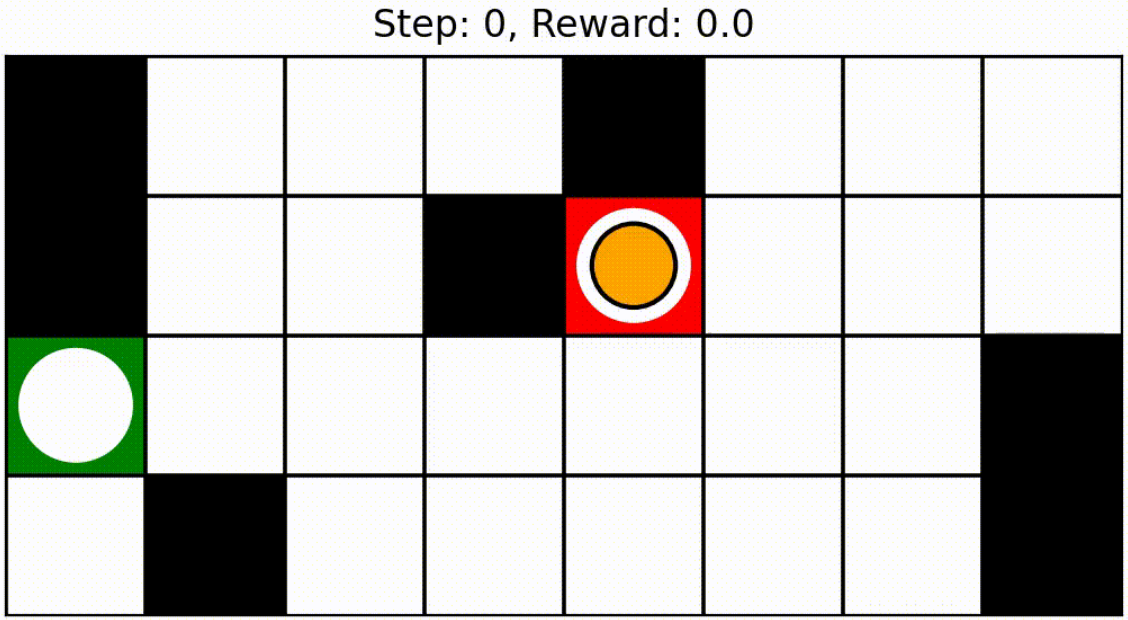
The Value Network



- Let's assign a point value of **2** for our goal and set positions on the map based on those points.
- This is a simplified representative example of what our value network might be trying to predict.



The networks learn optimal actions and values across many iterations.



Credit:
https://medium.com/@leo.damato/building-custom-grid-environments-for-reinforcement-learning-in-gymnasium-a-simple-guide-d433e27424ab

An even simpler RL example

- Consider being at a casino with 3 slot machines and unknown payout rates.
- You get 1,000 pulls.
- What should you do to maximize winnings?

- There is no true state here as the pulls are independent.
- The rewards will be learned by trying.
- Key tradeoff:
  - **Explore**: try different machines to gather information
  - **Exploit**: stick with the machine that seems best at present

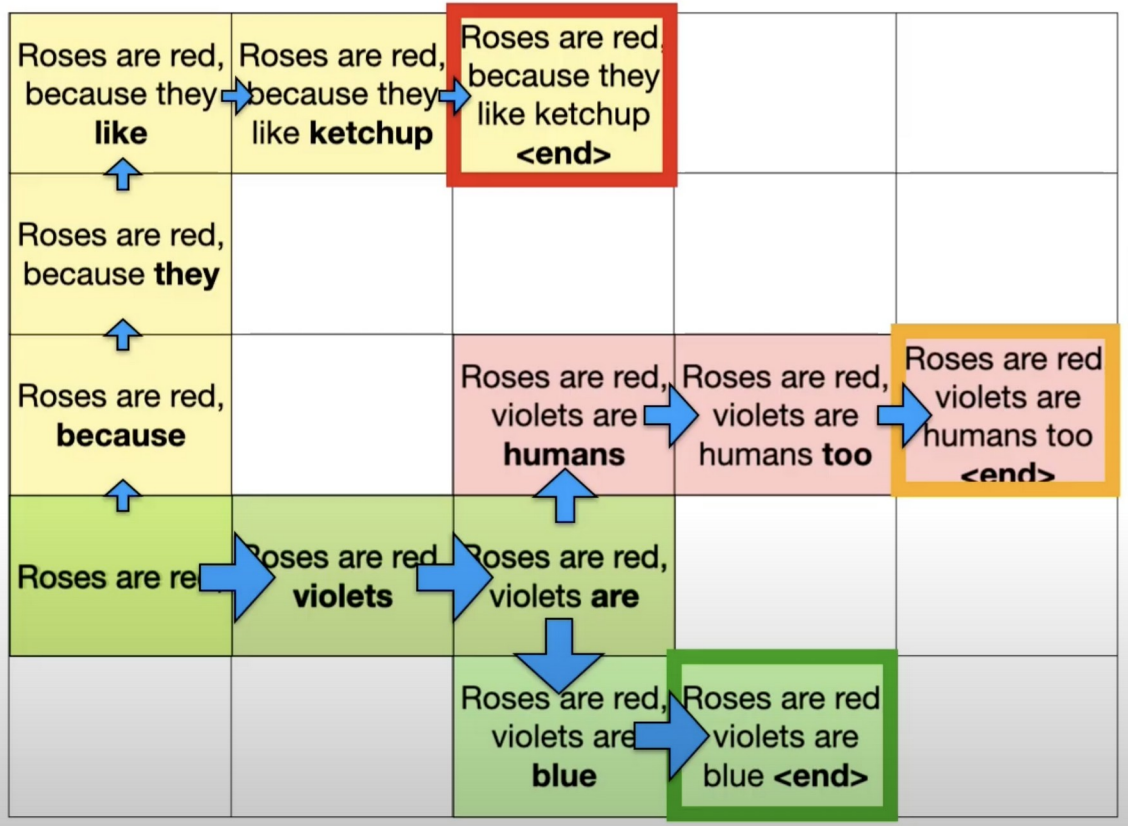
- **Epsilon-greedy strategy**: Pick the best machine most of the time while exploring ε% of the time.

Multi-Armed Bandit

- This is called the ”multi-armed bandit” problem and introduces a very basic form of reinforcement learning where the policy is a blend of exploitation with exploration.

- While it has no state in our construct, it has many business examples that do:
  - A/B testing or ad creative selection can employ exploit vs. explore tradeoffs, but have an effect on interactions and metrics like customer satisfaction.
  - This is why state is significant to reinforcement learning.
    - Relevant to our upcoming discussion of time series data.

Modern RL in Action: Training “human” bots



OpenAI trained a model to mimic human sounding text at scale!

(of course, it’s a high dimensional space, not a grid, but the idea holds)

Credit: Serrano Academy RLHF